HTE staff checklist - Setup

Edition 2026-10-11 | Local OT-2 | 60 first-round + 10 second-round wells

Campaign: _____ Operator/date: _____

Draft: 250 uL scale, 96-position mapping. A 24-vial rack needs revised protocols. Use released files after HTE confirms scale, glass geometry/seal, simultaneous heating of 60 reactions, pipetting/extraction/IS recovery and the 1.7-min LC method. Off-robot magnetic heating is requested; current files use slot 10. Release first dosing separately; calibration must pass before workup.

Before the clock

Check **59 pairs + one L17 reference**, all 30 active ligands covered, L07 excluded. Prepare stocks/reserves from `stock_preparation.csv`; prepare ligands in the glovebox. Label IDs, assays and addresses. Use plastic stock racks; metal blocks are for reactions. Keep reserves capped. Start the local watcher; key: laptop `anthropic.key`.

Stock	Preparation
SM + naphthalene	1,000 mM SM + 10 mM IS in toluene; at least 5.00 mL.
30 active ligands / Pd	20 mM ligands / 40 mM Pd(COD)(DQ), in toluene. CSV gives masses/reserves.
Toluene	Stock solvent; no extra neat dose in the default recipe.
Workup / LC diluent	1:1 water/ACN + 1% v/v HCOOH; no extra IS.

SM: PyFluor (pyridine-2-sulfonyl fluoride). **Product:** 2-fluoropyridine. Combined stock at 100% assay: **805.80 mg SM + 1.00 mL of 50 mM naphthalene/toluene**, toluene to **5.00 mL** in capped **10 mL glass/PTFE vial**. Naphthalene master: **12.817 mg to 2.00 mL**; separate toluene and LC-compatible masters.

Deck - use stage quantities from `deck_loads.csv`

Left: p300 single GEN2. **Right:** p300 eight-channel GEN2. Manual moves.

Slot	Contents
1 / 4	Ligand racks L01-L24 excluding L07 / L25-L31; exact CSV addresses.
2	Reservoir A1 SM/IS/toluene; A2 Pd; A3 unused by default; A4/A6 workup; A5/A7 LC diluent. Fill at most 11.9 mL/channel.
3 / 10	Reaction block for dosing/cooling / heater-shaker with released mount.
6	Round-two substrate array during dosing; LC plate during workup.
5 / 8 / 9	p300 single tips / p300 multi tips / p300 multi tips. Slots 7 and 11 empty.

Calibration - 14 injections during round one

Authentic SM/product: **100 mM, 2.00 mL each, no IS** (SM 32.232 mg; product 19.418 mg at 100% assay). Analyte working stock: **100 uL to 1.00 mL**, 10 mM. Analytical IS: **40 uL of 50 mM to 2.00 mL**, 1 mM.

For each analyte, prepare **1.00 mL per column**, adding **50 uL of 1 mM IS to every tube**. Use the approved matrix, accounting for stock solvents.

Analyte, uM	0 blank	100	500	1,000	2,500	6,000	5,000 check
Stock, mM	None	10	10	10	100	100	100
Analyte, uL	0	10	50	100	25	60	50
Matrix, uL	950	940	900	850	925	890	900

Fit blank plus five levels. Keep the independent 5,000 uM check outside the fit.

HTE staff checklist - Run

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Block/seal: _____ LC method/channel: _____

Follow App pauses and released CSV maps. Preserve orientation/sample IDs. Start the timer with first dosing; stocks/initial selection are prepared beforehand.

Step	Staff action
1. Dose	Import released `dose_and_react.py`. Check deck/offsets. Dose **25 uL of 1 M SM/10 mM IS in toluene into 60 wells**. Retain the solvent; proceed directly to catalyst dosing.
2. React	Pair: **75 uL ligand A + 75 uL ligand B + 75 uL Pd**. Reference: **150 uL L17 + 75 uL Pd**. With the retained 25 uL SM stock, total is **250 uL**. Seal/move to slot 10; **95 C / 500 rpm / 4 h**, with validated thermal lag.
3. During hold	Predose **25 uL SM/IS/toluene into ten round-two wells**. Cap promptly, label and store under the approved conditions; retain solvent. Resume promptly. Run **14 calibration injections**. Cap unused sources; uncover before aspiration.
4. Cool / release	Follow the thermal-handling pause; move sealed block off heater to slot 3 and cool before opening. After calibration passes, obtain the separately released first workup protocol and deck loads.
5. Work up	Load LC plate in slot 6. Run released workup: add **250 uL workup solvent, no IS**; move/shake **2 min without heating** and return as prompted. Confirm homogeneous extract; remix if delayed, clear bubbles.
6. LC	Fresh sample tips prewet twice in their own well. Robot doses **180 uL diluent + 20 uL extract**. Seal/mix LC plate **2 min**, clear bubbles, check needle height/volume. Run **60 samples** in the mapped sequence.
7. Decide	Normalize the complete Chromeleon export with injection-number mapping; publish completion marker. Wait for calibration/QC and **ten new pairs** in `next_dosing.csv`. Import the released second-round protocol.
8. Round two	Verify retained **25 uL predoses** and mapping. Refill sources from reserves. **Do not redose SM/IS or evaporate.** Add selected catalysts, seal and run the same four-hour hold. Cool and repeat workup/LC for **ten samples**. Stop the clock after analysis; archive results/QC/logs.

Pause and correct

- Validate **greater than 1 mm** bottom clearance and liquid immersion; sample default is 1.5 mm. Stop for contact, unintended air, dripping, bubbles or solids.
- Reuse only validated same-stock/noncontact dosing tips. Samples use fresh tips. Never return to source after LC contact. Uncertain transfer: fresh documented dilution, no guessed top-up.
- Two phases/solids: use approved extraction before sampling. Keep sealed through heating/cooling. Retain failed results; repeat for documented QC faults, never to choose a higher yield.

Per reaction: 25 umol SM, 250 uL, 0.1 M; 6 mol% each ligand, 12 mol% Pd; no base/external fluoride. Requested LC: **1.7 min**, 70 reaction samples. Provisional total **686 / 720 min**; measure cycle/handling. **Workflow/code due 21 Oct AoE.** Details: reference handbook.